NIFTY 50 Regime Dashboard

Project Report - everything we built

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An honest, leak-free forecasting dashboard for the NIFTY 50 index. It learns from 10 years of real 1-minute data and predicts the market REGIME at three horizons (intraday / weekly / monthly). The headline honesty principle: it only claims an edge where the data really supports one.

- Reliable signal: VOLATILITY (calm vs volatile) - ~58 to 77% accurate.
- Hard / near-random: DIRECTION (up / down / flat) - ~35%, shown but flagged weak.
- Built on honest walk-forward validation: no look-ahead, leakage-tested, overfitting-checked.

This report covers what the dashboard predicts, the seven improvements we added, the honest findings (with the exact live numbers), and how to run and update it.

1. What the dashboard predicts

Four models, each on real NIFTY 50 data, validated out-of-sample. 'Hit ratio' is how often the model's call was right on data it never trained on.

Model	Predicts	Hit ratio	vs baseline
Intraday 5-min	volatility (next ~4h)	77%	+20.8 pts
Intraday 15-min	volatility (next ~4h)	76%	+18.8 pts
Weekly	direction (5 days)	58%	+10.4 pts
Monthly	volatility (21 days)	74%	+17.3 pts

The honest core

- VOLATILITY is predictable because it clusters (a stormy day tends to follow a stormy day). This is the model's real edge.
- DIRECTION (will it go up or down) is close to a coin-flip at short horizons - the efficient-market wall. We show it but mark it LOW-trust so nobody over-trusts it.

2. The seven improvements we added

1. One-command refresh

A single script rebuilds every dashboard data file (13 steps, ~2 min) - was 8+ manual steps.

2. Confidence calibration

Makes the % trustworthy: a reliability diagram + isotonic fix so '70%' really means 70%.

3. HAR-RV benchmark

Added the HAR-RV econometric model; benchmarked it vs XGBoost / persistence / ensemble.

4. Live track record

Rolling out-of-sample scoreboard (all-time vs last-30 vs last-10) to catch model drift.

5. 'Today's read'

Plain-English summary that turns the live calls into one human sentence.

6. Equity-curve simulator

Turns the volatility signal into P&L: a risk-off overlay vs buy-and-hold.

7. Feature value test

Honest ablation: keep a new input only if it beats the simpler model out-of-sample.

3. The honest findings (live numbers)

Calibration - is the confidence honest?

Model	Shown vs real	Verdict	ECE raw -> cal
Intraday 5m	under confident	under-confident	0.100 -> 0.003
Intraday 15m	under confident	under-confident	0.088 -> 0.005
Weekly	over confident	over-confident	0.084 -> 0.044
Monthly	under confident	under-confident	0.077 -> 0.031

Finding: intraday/monthly were UNDER-confident (better than they claimed); weekly a bit over-confident. Calibration cuts the error (ECE) to near zero.

Volatility model benchmark

5d: HAR-RV 65% XGBoost 65% Persistence 65% Ensemble 65% (best: HAR-RV 65%, baseline 58%)

21d: HAR-RV 61% XGBoost 57% Persistence 69% Ensemble 57% (best: Persistence 69%, baseline 58%)

HAR-RV is competitive weekly; the simple persistence rule wins monthly. Ensembling did not beat them - don't over-engineer.

Does the volatility signal pay? (risk-off overlay vs buy-and-hold)

Metric	Buy & hold	Vol overlay
Total return	171%	53%
CAGR	18.3%	7.4%
Annual volatility	15.0%	9.1%
Sharpe	1.2	0.83
Max drawdown	-17.2%	-10.8%

The overlay cuts drawdown and volatility (the signal is real) but gave up too much upside in a strong bull run - best for risk control, not a naive on/off switch.

4. More findings & how to run it

Feature value test (overnight gap vs intraday range)

5d: 65.0%, + overnight gap 65.3%, + gap + range 65.2%

21d: 73.2%, + overnight gap 74.0%, + gap + range 73.2%

Overnight gap adds a small, consistent edge; intraday range is noise. Every new input is vetted this way - kept only if it earns its place.

Live track record (recent vs all-time)

Model	All-time	Last 30	Last 10
Intraday 5-min	75.4%	70.0%	90.0%
Intraday 15-min	76.2%	73.3%	80.0%
Weekly	57.9%	66.7%	50.0%
Monthly	73.6%	90.0%	70.0%

If 'Last 30' falls well below 'All-time', the model may be drifting - a signal to retrain. Right now all four are holding up.

How to run & update

- Rebuild everything: python refresh_dashboard.py (add --retrain to retrain models).
- Deployed live from the remine-ml repo's /docs folder (GitHub Pages).
- To go fully current, append newer NIFTY 1-minute data, then rerun refresh.

Bottom line: an honest, self-updating regime dashboard. It is upfront about what it can (volatility) and cannot (direction) predict, proves its calibration, tracks its own live accuracy, and shows how the edge translates to risk and return.